



Hospital Engineering and Management

Lecture 9: Hospital Pharmacy Services – Planning, Medication Safety and Engineering Support

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Course Code: UBM702

Learning Objectives

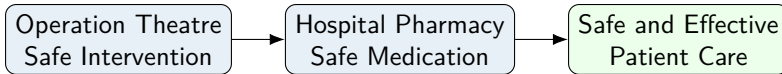
By the end of this lecture, students should be able to:

- ▶ explain the role and organization of hospital pharmacy services;
- ▶ trace the medication-use process from selection and procurement to administration and monitoring;
- ▶ apply basic principles of pharmacy layout, storage, inventory control and cold-chain management;
- ▶ identify engineering and information-system requirements that support safe medication delivery;
- ▶ recognize common medication-safety risks and system controls;
- ▶ describe the biomedical engineer's role in pharmacy equipment, alarms, automation and quality assurance.

Connection with the Previous Lecture

Lecture 8 Established

The operation theatre is a tightly controlled environment where reliable utilities, sterile workflow, equipment and human coordination directly affect surgical safety.



Key Shift

Medication safety depends on much more than the drug itself: selection, storage, identification, dispensing, delivery, administration and monitoring must all work correctly.

Opening Question: How Does a Tablet Reach the Right Patient?

Suppose a physician orders an antibiotic for a patient. What must happen before the patient receives the first dose?

- ▶ correct medicine must be available;
- ▶ prescription must be clear and complete;
- ▶ allergy and dose must be checked;
- ▶ medicine must be selected correctly;
- ▶ label must identify drug and patient;
- ▶ storage conditions must have been maintained;
- ▶ medicine must reach the correct ward;
- ▶ administration must be documented and monitored.

A medication error can originate anywhere in the medication-use system.

What Is a Hospital Pharmacy?

Definition

A hospital pharmacy is the service responsible for the safe selection, procurement, storage, preparation, dispensing and management of medicines used for patient care within the hospital.

It links clinical decisions with a controlled supply system.

- ▶ provides medicines to inpatients, outpatients and emergency areas;
- ▶ supports medication review and patient safety;
- ▶ maintains inventory and storage conditions;
- ▶ manages controlled and high-risk medicines according to applicable policies;
- ▶ coordinates with clinicians, nursing, stores, finance and biomedical engineering.

Core Objectives of Hospital Pharmacy Services

Clinical Objectives

- ▶ right medicine for the right patient;
- ▶ appropriate dose, route and timing;
- ▶ prevention of interactions and duplication;
- ▶ support for rational medication use.

Operational Objectives

- ▶ continuous availability;
- ▶ safe storage;
- ▶ traceability;
- ▶ minimum avoidable expiry and wastage;
- ▶ efficient distribution.

Clinical safety + supply reliability + traceability = effective pharmacy service

Hospital Pharmacy within the Hospital System



Systems View

The pharmacy is both a clinical service and a logistics system. Its performance depends on information flow, material flow, environmental control and human verification.

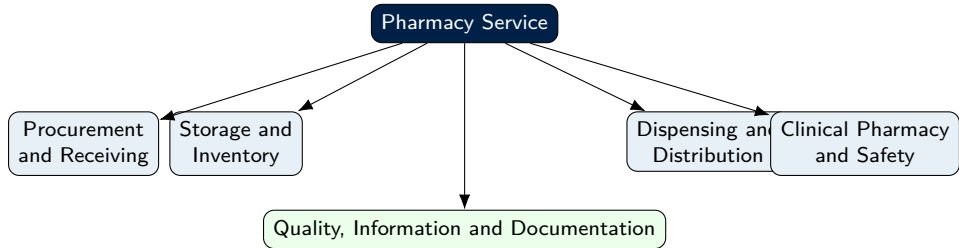
Common Pharmacy Service Models

- ▶ **Central pharmacy:** major procurement, storage and dispensing functions concentrated at one location.
- ▶ **Satellite pharmacy:** smaller pharmacy unit near a high-demand clinical area such as ICU, OT or emergency.
- ▶ **Ward stock system:** selected medicines stored in controlled ward locations for immediate access.
- ▶ **Unit-dose distribution:** individually packaged doses supplied for a specific patient and administration time.
- ▶ **Automated dispensing:** secured cabinets or systems provide controlled access and electronic tracking.

Design Principle

The distribution model should balance speed of access with medication safety, inventory control and traceability.

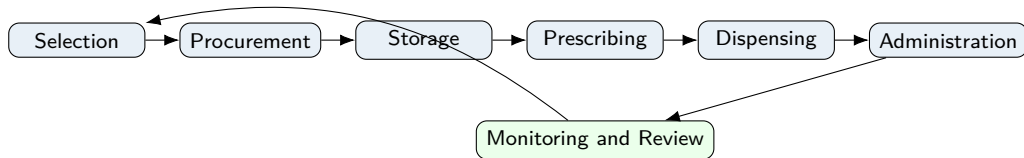
Functional Organization of a Hospital Pharmacy



People and Responsibilities

- ▶ **Pharmacist:** validates orders, dispenses medicines, counsels patients and supports medication safety.
- ▶ **Clinical pharmacist:** reviews therapy, interactions, duplication, dose adjustment and monitoring needs.
- ▶ **Pharmacy technician / assistant:** supports stock handling, preparation, packing and documentation under supervision.
- ▶ **Nurse:** receives, stores locally, administers and documents medicines at the point of care.
- ▶ **Biomedical / facility engineer:** maintains refrigerators, cabinets, balances, clean-air equipment, alarms and backup systems.
- ▶ **IT team:** maintains electronic prescribing, pharmacy information system, barcode and interfaces.

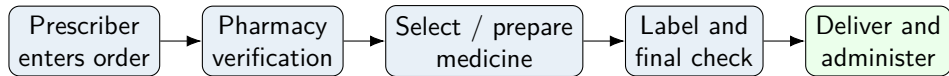
The Medication-Use Cycle



Engineering Insight

Every stage needs controls: identifiers, environmental monitoring, alarms, access control, data integrity and reliable equipment.

Prescription and Medication-Order Flow



- ▶ patient identity, medicine, dose, route and timing must be clear;
- ▶ allergies and relevant clinical information should be available;
- ▶ ambiguous or unsafe orders require clarification before dispensing;
- ▶ electronic systems can reduce transcription steps but still require verification.

Medicine Selection and the Hospital Formulary

Formulary Concept

A hospital formulary is an approved list of medicines selected for routine use based on clinical need, safety, effectiveness, quality and resource considerations.

Typical benefits include:

- ▶ standardization of commonly used medicines;
- ▶ reduced unnecessary duplication;
- ▶ improved procurement and stock planning;
- ▶ easier development of treatment and safety protocols;
- ▶ better staff familiarity with available strengths and formulations.

Pharmacy and Therapeutics Committee

A multidisciplinary committee may include clinicians, pharmacists, nursing, administration and other experts.

Typical Responsibilities

- ▶ formulary decisions;
- ▶ medication-use policies;
- ▶ high-risk medicine controls;
- ▶ review of medication incidents;

Quality Functions

- ▶ antimicrobial stewardship support;
- ▶ prescribing audits;
- ▶ standard concentrations;
- ▶ medication-safety improvement.

Procurement: More Than Buying Medicines

Safe procurement requires control of:

- ▶ approved suppliers and product specifications;
- ▶ batch, expiry and quantity verification;
- ▶ transport conditions where temperature-sensitive products are involved;
- ▶ recall information and traceability;
- ▶ stock rotation and reorder planning;
- ▶ documentation of receipt, rejection and return.

Risk Example

A product may be clinically correct but unusable if its storage history is unknown or if packaging integrity is compromised.

Basic Inventory-Control Questions

For every important medicine, the pharmacy should know:

- ▶ how much is currently available?
- ▶ how fast is it being consumed?
- ▶ how long does replenishment take?
- ▶ what minimum safety stock is needed?
- ▶ when should a new order be placed?
- ▶ which items are expensive, critical or fast-moving?

Simple Reorder Concept

Reorder level $\approx$ Expected use during lead time + Safety stock

ABC and VED Thinking

ABC Analysis

Classifies items by annual consumption value.

- ▶ A: high financial impact;
- ▶ B: intermediate;
- ▶ C: lower financial impact.

VED Analysis

Classifies items by clinical criticality.

- ▶ V: vital;
- ▶ E: essential;
- ▶ D: desirable.

Management Insight

An inexpensive but vital emergency medicine may deserve tighter stock control than a costly but non-critical item.

Storage Zones in a Hospital Pharmacy

A well-planned pharmacy separates functions logically.

- ▶ receiving and quarantine area;
- ▶ general medicine storage;
- ▶ refrigerated / cold-chain storage;
- ▶ high-alert or restricted-access storage;
- ▶ flammable or special-hazard storage where applicable;
- ▶ dispensing and checking counters;
- ▶ returned / recalled / expired stock area;
- ▶ records, staff and counselling areas.

Design Principle

Usable stock should never be confused with returned, damaged, recalled or expired stock.

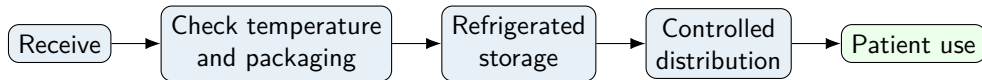
Storage Environment and Monitoring

Medicine quality can be affected by temperature, moisture, light and contamination.

- ▶ temperature monitoring;
- ▶ humidity control where required;
- ▶ protection from direct sunlight;
- ▶ adequate ventilation;
- ▶ pest and water-ingress prevention;
- ▶ shelving off the floor;
- ▶ clear labelling and segregation;
- ▶ calibrated monitoring devices;
- ▶ alarm and escalation process;
- ▶ documented corrective action.

Cold-Chain Management

Some medicines and biological products require controlled refrigeration throughout storage and transport.



Critical Engineering Requirement

Refrigeration failure must be detected quickly, and there must be a documented response for product relocation and assessment.

Designing a Reliable Pharmacy Refrigerator System

A hospital pharmacy refrigerator should be supported by:

- ▶ dedicated and clearly identified electrical supply;
- ▶ emergency / backup power where continuity is essential;
- ▶ continuous or frequent temperature logging;
- ▶ local and remote alarm where appropriate;
- ▶ calibrated temperature sensor;
- ▶ preventive maintenance and condenser cleaning;
- ▶ defined contingency location for stock transfer.

Refrigerator + sensor + alarm + backup power + response plan = cold-chain reliability

High-Alert Medicines

Some medicines can cause particularly serious harm if used incorrectly.

Risk reduction commonly includes:

- ▶ restricted access;
- ▶ standardized concentrations where feasible;
- ▶ prominent identification;
- ▶ independent double-checks for selected steps;
- ▶ smart-pump drug libraries for compatible infusions;
- ▶ clear separation from similar-looking products;
- ▶ incident review and staff education.

Key Principle

For high-risk processes, do not rely on memory alone. Build multiple system barriers.

Look-Alike / Sound-Alike (LASA) Risk

Medication errors can occur when names or packages are easily confused.

Contributing Factors

- ▶ similar brand or generic names;
- ▶ similar packaging;
- ▶ poor lighting;
- ▶ rushed selection;
- ▶ adjacent shelf placement.

System Controls

- ▶ physical separation;
- ▶ shelf alerts;
- ▶ barcode verification;
- ▶ clear typography;
- ▶ final independent check.

Unit-Dose Distribution

In a unit-dose approach, medication is supplied in patient-specific, ready-to-administer quantities whenever appropriate.

Potential advantages:

- ▶ fewer handling steps on the ward;
- ▶ improved patient identification and labelling;
- ▶ better traceability of doses;
- ▶ reduced ward stock and uncontrolled access;
- ▶ easier return and reconciliation of unused doses.

Engineering Connection

Unit-dose systems may use packaging machines, barcode printers, carts and automated storage that require maintenance and validation.

Ward Stock and Emergency Stock

Ward stock allows rapid access to selected medicines, but uncontrolled stock increases risk. Good control requires:

- ▶ approved list and maximum quantities;
- ▶ secure storage;
- ▶ regular expiry checking;
- ▶ temperature control where required;
- ▶ documented replenishment;
- ▶ separation of external-use and internal-use products where relevant;
- ▶ clear accountability for high-risk items.

Automated Dispensing Cabinets

Automated dispensing cabinets can provide secure, electronically tracked access to medicines near patient-care areas.

Possible Benefits

- ▶ controlled access;
- ▶ electronic inventory;
- ▶ faster availability;
- ▶ user-level audit trail;

New Risks

- ▶ incorrect stocking;
- ▶ override misuse;
- ▶ network failure;
- ▶ drawer / sensor faults;
- ▶ data-interface mismatch.

Inpatient and Outpatient Pharmacy: Different Flows

Inpatient Pharmacy

- ▶ patient-specific ward supply;
- ▶ coordination with nursing administration times;
- ▶ returns and reconciliation;
- ▶ urgent / STAT medication support.

Outpatient Pharmacy

- ▶ prescription reception;
- ▶ queue management;
- ▶ dispensing and payment interface;
- ▶ patient counselling and take-home labelling.

Planning the Pharmacy Location

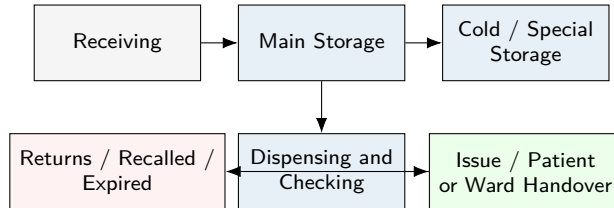
Pharmacy location should consider proximity to:

- ▶ outpatient flow for public dispensing;
- ▶ inpatient vertical transport / service lifts;
- ▶ emergency and critical-care areas;
- ▶ stores and receiving dock;
- ▶ secure delivery routes;
- ▶ information and communication infrastructure.

Planning Conflict

The pharmacy must be accessible to care areas but protected from uncontrolled public movement and unauthorized access.

A Simple Pharmacy Layout Concept



Flow Principle

Receiving, usable storage, dispensing and rejected / returned material should follow clearly controlled paths.

Engineering Services Required in a Pharmacy

- ▶ reliable electrical supply and adequate sockets;
- ▶ emergency power for critical refrigerators and information systems;
- ▶ suitable lighting for label reading and checking;
- ▶ HVAC appropriate to storage and work areas;
- ▶ network connectivity and secure information systems;
- ▶ fire detection and appropriate fire protection;
- ▶ access control and CCTV where policy requires;
- ▶ water and drainage for designated preparation / cleaning functions.

Pharmacy Information System

A pharmacy information system can support:

- ▶ patient medication profiles;
- ▶ electronic order review;
- ▶ stock and expiry management;
- ▶ batch and recall tracking;
- ▶ billing and charge capture;
- ▶ interaction / allergy alerts;
- ▶ dispensing records;
- ▶ reports and quality indicators.

Important

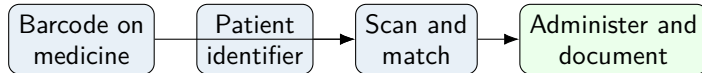
Automation does not eliminate error. Poor data, wrong configuration or interface failures can create new error pathways.

Electronic Prescribing and CPOE

Computerized provider order entry can reduce handwriting and transcription problems. However, safe implementation still requires:

- ▶ correct patient selection;
- ▶ appropriate dose and route choices;
- ▶ meaningful allergy and interaction alerts;
- ▶ good screen design;
- ▶ prevention of duplicate orders;
- ▶ downtime procedures;
- ▶ audit of alert overrides and unusual patterns.

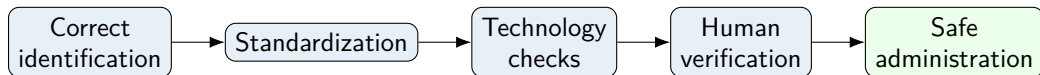
Barcode-Enabled Medication Workflow



Purpose

Barcode systems add an identification check between the ordered medicine, the prepared dose and the intended patient.

Medication Safety as an Engineered System



Safety Philosophy

Good systems make the correct action easier, the incorrect action harder, and failures easier to detect before reaching the patient.

Common Dispensing-Error Pathways

Examples include:

- ▶ wrong patient selected;
- ▶ wrong medicine or strength picked;
- ▶ decimal-point or dose interpretation error;
- ▶ wrong route or dosage form;
- ▶ incorrect label;
- ▶ expired product selected;
- ▶ allergy or interaction overlooked;
- ▶ similar packaging mistaken;
- ▶ incomplete handover or documentation.

Question for Students

Which of these can be prevented by layout, barcode, software, storage design or automation?

Infusion Systems and Smart Pumps

Many hospital medicines are delivered using infusion pumps.

- ▶ volumetric pumps control fluid delivery rate;
- ▶ syringe pumps are used for small-volume precise infusions;
- ▶ programmable drug libraries can provide dose-rate limits;
- ▶ alarm settings must be clinically meaningful;
- ▶ drug-library updates require governance and version control;
- ▶ pump performance must be tested and maintained.

Interface Risk

A correct medicine can still harm a patient if the infusion rate, concentration or pump programming is wrong.

Biomedical Engineer's Role in Pharmacy Services

The biomedical engineer may support:

- ▶ medicine refrigerators and temperature alarms;
- ▶ infusion and syringe pumps;
- ▶ balances and measurement devices;
- ▶ clean-air / containment equipment used for preparation;
- ▶ automated dispensing and packaging systems;
- ▶ barcode printers and scanners with IT support;
- ▶ environmental monitoring sensors;
- ▶ preventive maintenance, calibration and electrical-safety testing.

Maintenance Priorities for Pharmacy Equipment

High Priority

- ▶ refrigerators / freezers;
- ▶ infusion devices;
- ▶ clean-air equipment;
- ▶ automated dispensing systems;
- ▶ temperature / humidity monitoring.

Typical Checks

- ▶ accuracy;
- ▶ alarms;
- ▶ power backup;
- ▶ filters and airflow;
- ▶ batteries;
- ▶ communication interfaces.

Sterile Medication Preparation: Basic Concept

Some medicines require preparation in a controlled environment to reduce contamination risk. Key design ideas include:

- ▶ controlled work zone;
- ▶ appropriate hand hygiene and gowning;
- ▶ cleanable surfaces;
- ▶ controlled airflow where required;
- ▶ separation of preparation from general traffic;
- ▶ documented cleaning and environmental monitoring;
- ▶ equipment qualification and maintenance.

Boundary of This Lecture

Detailed sterilization of instruments and the Central Sterile Supply Department will be covered in the next lecture.

Clean-Air and Containment Equipment

Depending on the preparation task, specialized cabinets may be required.

Engineering concerns include:

- ▶ airflow direction and velocity;
- ▶ HEPA filtration where applicable;
- ▶ filter loading and replacement;
- ▶ cabinet alarms;
- ▶ room pressure relationships;
- ▶ certification / performance verification;
- ▶ safe exhaust or containment for hazardous tasks.

Do Not Confuse Functions

Product protection, worker protection and environmental protection are different objectives and require appropriate equipment selection.

Hazardous Medication Handling

Some medicines require additional controls to protect staff and the environment. General risk-reduction principles include:

- ▶ restricted and clearly identified preparation areas;
- ▶ suitable engineering controls;
- ▶ personal protective equipment;
- ▶ spill-management procedures;
- ▶ closed or contained transfer techniques where appropriate;
- ▶ dedicated waste pathway;
- ▶ staff training and exposure-response procedures.

Returns, Recalls and Expired Medicines

Returned, recalled or expired medicines should be physically and electronically segregated from usable stock.

- ▶ record reason for return;
- ▶ prevent accidental re-dispensing;
- ▶ identify affected batch or lot where relevant;
- ▶ trace patients or locations if recall action is required;
- ▶ follow authorized disposal / return procedures;
- ▶ analyze repeated expiry to improve inventory planning.

Emergency Medicines and Crash Carts

Emergency medicines must be immediately available, standardized and reliably maintained.

- ▶ standardized cart layout;
- ▶ tamper-evident control where used;
- ▶ defined responsibility for restocking;
- ▶ regular expiry checks;
- ▶ resuscitation drugs grouped logically;
- ▶ compatibility with emergency equipment workflow;
- ▶ clear documentation after use.

Engineering Connection

Emergency medication access is inseparable from defibrillators, suction, oxygen, monitors and emergency power.

Antimicrobial Stewardship: Pharmacy Contribution

Pharmacy data and clinical review can support responsible antimicrobial use through:

- ▶ review of selected prescriptions;
- ▶ dose adjustment and duration review;
- ▶ antibiotic-consumption monitoring;
- ▶ formulary restrictions where adopted;
- ▶ collaboration with microbiology and infection-control teams;
- ▶ feedback to prescribers and clinical units.

Systems Link

Laboratory results, pharmacy data and clinical decisions must connect to improve antimicrobial use.

Useful Pharmacy Performance Indicators

Examples of measurable indicators include:

- ▶ prescription-to-dispensing turnaround time;
- ▶ stock-out frequency;
- ▶ value of expired / wasted stock;
- ▶ cold-chain excursions;
- ▶ medication dispensing incidents;
- ▶ percentage of orders requiring clarification;
- ▶ inventory accuracy;
- ▶ emergency medicine availability;
- ▶ preventive-maintenance compliance for critical pharmacy equipment.

Failure Mode Thinking in Pharmacy

Failure	Possible Effect	System Control
Refrigerator failure	Product temperature excursion	alarm, backup power, contingency storage
Wrong shelf selection	Wrong medicine dispensed	separation, barcode, final check
Network outage	Delay / missing electronic orders	downtime procedure, reconciliation
Expired stock remains active	Unsafe / ineffective product use	FEFO, expiry alerts, segregation
Pump library mismatch	Incorrect dose-rate guardrails	controlled update and validation

Case Study 1: Pharmacy Refrigerator Failure

At 02:00, the main pharmacy refrigerator fails. Temperature begins to rise while temperature-sensitive medicines are inside.

Discussion Questions

1. How should the failure be detected?
2. Who should receive the alarm?
3. Where should the medicines be moved?
4. How should affected stock be identified and documented?
5. Which engineering checks should be performed before returning the refrigerator to service?

Lesson

Equipment maintenance alone is insufficient; a safe system also requires monitoring, escalation and contingency planning.

Case Study 2: Similar Packages, Wrong Strength

Two strengths of the same medicine have nearly identical packaging and are stored side by side. A busy staff member selects the wrong strength.

Immediate Causes

- ▶ visual similarity;
- ▶ adjacent storage;
- ▶ time pressure;
- ▶ inadequate final verification.

System Improvements

- ▶ separate locations;
- ▶ shelf alert labels;
- ▶ barcode scan;
- ▶ independent check for high-risk doses.

Case Study 3: Electronic Order, Wrong Patient

A prescriber opens the wrong patient's chart and enters a medication order. The electronic system transmits it correctly to pharmacy.

Important Insight

A technically functioning information system can still propagate a human identification error very efficiently.

Discussion:

- ▶ what interface cues could reduce wrong-patient selection?
- ▶ where can pharmacy detect the mismatch?
- ▶ how can barcode administration add another barrier?
- ▶ what should be audited after such an event?

Classroom Activity: Design a 300-Bed Hospital Pharmacy

Work in small groups and propose a functional pharmacy plan for a 300-bed hospital. Include:

1. receiving and quarantine;
2. general storage;
3. cold-chain storage;
4. dispensing / checking;
5. inpatient and outpatient issue points;
6. returns / expired stock segregation;
7. one automated system;
8. backup power and alarm strategy.

Deliverable: one block diagram + five design justifications.

Classroom Exercise: ABC-VED Reasoning

Consider four medicine groups:

Item	Annual Cost Impact	Clinical Criticality
A	High	Vital
B	High	Desirable
C	Low	Vital
D	Low	Desirable

Question

Which items require the tightest management attention, and why? Explain from both financial and patient-safety perspectives.

Mini-Project Ideas Related to Hospital Pharmacy

- ▶ IoT-based temperature monitoring and alarm for pharmacy refrigerators;
- ▶ RFID / barcode medication inventory prototype;
- ▶ smart expiry-tracking dashboard using FEFO logic;
- ▶ automated medicine-dispensing cabinet concept;
- ▶ infusion-pump maintenance and alarm-management audit;
- ▶ pharmacy layout optimization using patient and material flow;
- ▶ medication cold-chain risk assessment using FMEA;
- ▶ dashboard for stock-out, expiry and turnaround-time indicators.

What Should a Biomedical Engineer Observe During a Pharmacy Visit?

Look beyond shelves of medicines. Observe:

- ▶ refrigerator power source and temperature logging;
- ▶ alarm visibility and escalation route;
- ▶ condition of infusion pumps and charging areas;
- ▶ barcode / printer / scanner workflow;
- ▶ network and downtime arrangements;
- ▶ clean-air or containment equipment, if present;
- ▶ environmental monitoring;
- ▶ equipment labels, calibration status and preventive-maintenance records.

Key Takeaways

- ▶ Hospital pharmacy is a clinical, logistical and information-intensive service.
- ▶ Medication safety depends on the complete medication-use cycle, not only on dispensing.
- ▶ Layout, segregation, storage environment and cold-chain reliability directly affect medicine quality.
- ▶ Barcodes, electronic prescribing and automated dispensing add safety barriers but can introduce new technical risks.
- ▶ High-alert and look-alike / sound-alike medicines require stronger system controls.
- ▶ Biomedical engineers support pharmacy safety through reliable equipment, alarms, environmental monitoring and maintenance.

One-Slide Summary



Right medicine + right information + right environment + right system = safer medication use

Assignment for the Next Lecture

Visit or study the hospital pharmacy of any hospital and prepare a one-page engineering observation note.

Include:

1. pharmacy service model: central, satellite or mixed;
2. a simple block layout;
3. method of cold-chain monitoring;
4. method of inventory / expiry tracking;
5. one medication-safety barrier;
6. three pieces of equipment requiring biomedical / engineering support;
7. one realistic improvement suggestion.

Suggested Reading

- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ World Health Organization publications on medication safety and safe medication practices.
- ▶ Hospital medication-management policies and applicable national accreditation / regulatory guidance.

Next Lecture: Central Sterile Supply Department (CSSD)

Thank You

Questions and Discussion